

Fig. 8.10

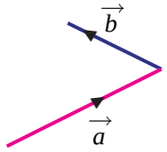


Fig. 8.11

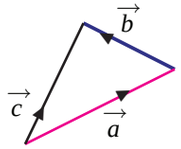


Fig. 8.12

Bringing the initial point of $\vec{b}$ to the terminal point of $\vec{a}$ (Fig. 8.11), we can get the resultant of these two forces (see Fig. 8.12). This motivates us to define the sum of two vectors.

Triangle law of addition

Let $\vec{a}$ and $\vec{b}$ be two vectors. Let A_1 and B_1 be the initial points of $\vec{a}$ and $\vec{b}$, and A_2 and B_2 be the terminal points of $\vec{a}$ and $\vec{b}$ respectively.

Draw A_3B_3 parallel to B_1B_2 so that $A_3B_3 = B_1B_2$. Then the vector $\overrightarrow{A_1B_3}$ is defined as the sum of the vectors $\vec{a}$ and $\vec{b}$, and it is denoted as $\vec{a} + \vec{b}$. This can be restated as,

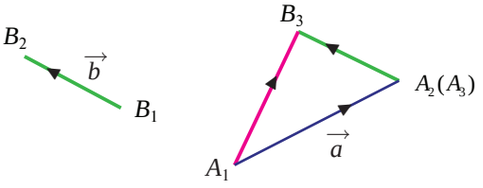


Fig. 8.13

Definition 8.10 (Triangle law of addition)

If two vectors are represented in magnitude and direction by the two sides of a triangle taken in order, then their sum is represented by the third side taken in the reverse order.

Result 8.1

If $\vec{a}, \vec{b}$ and $\vec{c}$ are the sides of a triangle taken in order then $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

Proof

Let $\overrightarrow{AB} = \vec{a}, \overrightarrow{BC} = \vec{b}$, and $\overrightarrow{CA} = \vec{c}$.

Now $\vec{a} + \vec{b} + \vec{c} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \overrightarrow{AC} + \overrightarrow{CA} = \overrightarrow{AA} = \vec{0}$.

Thus the result is proved.

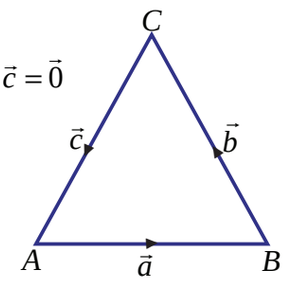


Fig. 8.14

Parallelogram law of vector addition

Let $\vec{a}$ and $\vec{b}$ be two vectors. Assuming that the initial points of the two vectors are the same, let us find the sum according to Definition 8.7. Let A and B be the terminal points of $\vec{a}$ and $\vec{b}$ respectively (Fig. 8.15). To find $\vec{a} + \vec{b}$, we draw AC parallel to OB so that $OB = AC$ and declare that $\overrightarrow{OC}$ is the sum (Fig. 8.16). We observe that OA and BC are parallel (Fig. 8.17).

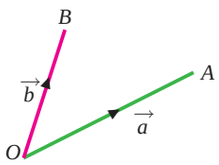


Fig. 8.15

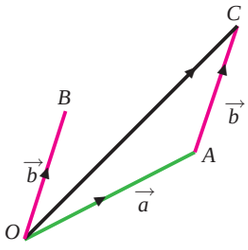


Fig. 8.16

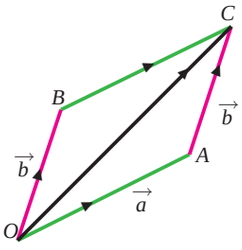


Fig. 8.17

So to find the sum of two vectors with the same initial point, draw the parallelogram with the given vectors as adjacent sides and declare the diagonal as the sum. Even the vectors do not have the

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